

TrES-3b

R-2831

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_150809 · created 2026-09-05T23:33:25+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457243.6946 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.057
Transit depth (Rp/Rs)^2	3.46 +/- 2.11 [%]
Orbital Inclination (inc)	83.57 +/- 2.92
Airmass coefficient 1 (a1)	22.95 +/- 14.39
Airmass coefficient 2 (a2)	-3.0 +/- 1.27
Scatter in the residuals of the lightcurve fit is	63.58 %
Best Comparison Star	#2 - [365, 244]
Optimal Aperture	1.22
Optimal Annulus	3.0
Transit Duration (day)	0.064 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.186 ± 0.057 vs 0.16309 (deviation 0.4σ)
Duration fit vs published	0.064 d vs 0.0567 d (deviation 0.33σ)
Mid-time O–C	1.2 min
Depth SNR	3.3
Residual scatter	63.58 %

Light curve

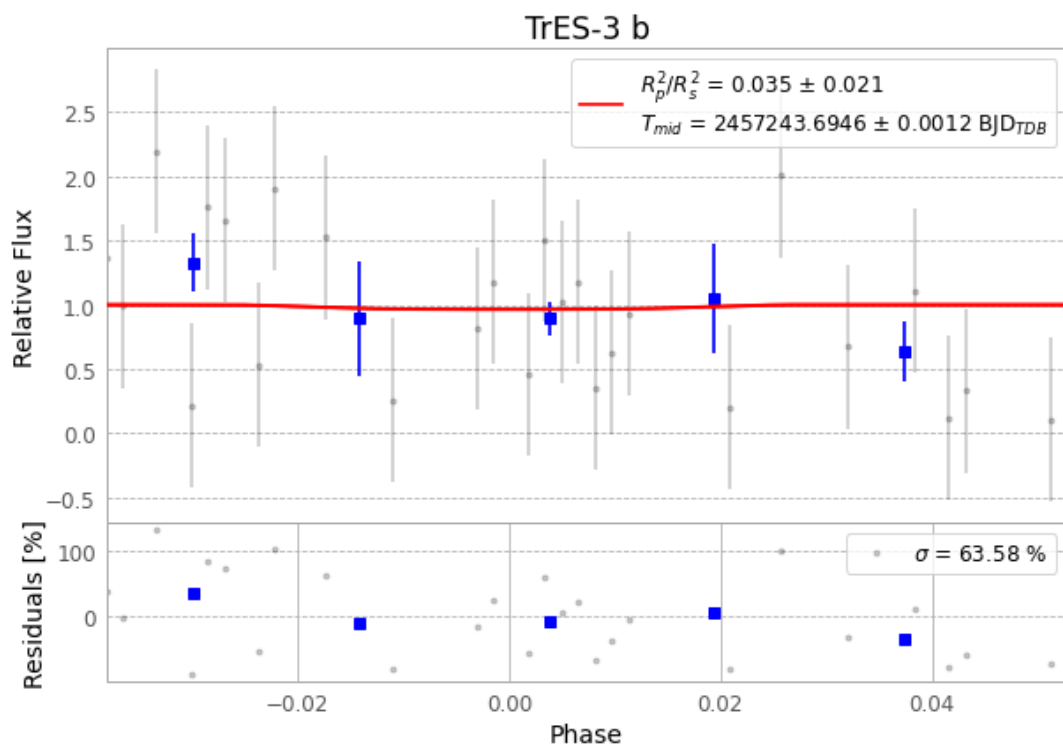


Figure 1 — FinalLightCurve_TrES-3 b_2015-08-08.png (SHA-256 59b515c5199bb758...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [375, 259], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0be543ca1db4feaa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

58 FITS frames (38 MB), TRES-3150809032415.FITS → TRES-3150809061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-150809.log (8bb5417d23ac...), FinalLightCurve_TrES-3 b_2015-08-08.png (59b515c5199b...), FinalLightCurve_TrES-3 b_2015-08-08.csv (f9497dbb6752...)

Compute resources

Wall time 126.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 23:33Z.